maquette-gltf

Render glTF 2.0 assets in Typst

github.com/bernsteining/maquette · bernsteining.github.io/maquette

Version 0.1.0 · August 26, 2026

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Introduction

maquette-gltf extends `maquette` with glTF 2.0 support: `.glb` or `.gltf` in, rendered frame out. The plugin adds a PBR pipeline (Cook-Torrance GGX + Karis split-sum IBL + WBOIT translucency) on top of maquette's shared rasterizer, plus glTF-specific machinery for authored cameras/lights/materials/animations and the ecosystem's compression + texture extensions.

This document only covers what maquette-gltf **adds**. The shared rendering knobs — camera framing, background, shadows, ground plane, SSAO/FXAA/SSAA, tone mapping — are already documented in maquette's manual and behave identically here. `render-gltf` accepts the same option dicts.

Quickstart

Two input shapes, both handled transparently.

`.glb` or fully-embedded `.gltf` — pass the bytes directly. This is the common case (Damaged Helmet, boombox, most Sketchfab downloads).

```
#import "@preview/maquette-gltf:0.1.0": render-gltf

#render-gltf(read("helmet.glb", encoding: none))
```

Split `.gltf` — the `.gltf` JSON references external `.bin` + textures by relative URI. Pass the **path string** plus an inline `read:` lambda; the wrapper walks the JSON, discovers every URI, reads each one through your lambda, packs them into a sidecar bundle for the plugin. You write zero filenames.

```
#render-gltf("Duck.gltf", read: p => read(p, encoding: none))
```

The `read:` lambda has to be an **inline lambda**, not a bare `read` reference. Typst resolves `read()` paths against the source file the call is textually in — a bare reference stays bound to the package's own path context. Wrapping in `p => read(p, ...)` gives the wrapper a filesystem handle scoped to your `.typ`.

Shared rendering config

Camera framing (`camera` / `center` / `up` / `fov` / `azimuth` / `elevation`), `background`, `shadows`, `ground`, `ssao`, `antialias`, `tone_mapping` — all inherited from maquette and behave the same on `render-gltf`. Don't repeat the tour here; refer to that manual.

One glTF-only knob to note: **`camera_auto_use: false`**. If the loaded asset ships an authored camera, `render-gltf` uses it by default (glTF viewer convention). Set this to `false` to force your Cartesian/spherical arguments to win instead.

```
#render-gltf(helmet, read: R,
  camera_auto_use: false,
  camera: (0, 0, 3), up: (0, 1, 0), fov: 60,
)
```

Image-Based Lighting

The dominant visual lever for PBR: an environment map lights the whole scene. Enabling `ibl` without an `hdr` bytes payload falls back to a procedural hemispheric sky.

Procedural IBL

```
#render-gltf(helmet, read: R,
  camera: (2.5, 1.5, 2.5), up: (0, 1, 0), fov: 40,
  background: "#181820",
  ibl: (intensity: 1.2),
)
```

HDR environment

Pass Radiance `.hdr` bytes as `ibl.hdr`. The prefilter builds a diffuse irradiance map + a specular mip chain up-front (once per HDR, cached).

```
#render-gltf(helmet, read: R,
  ibl: (intensity: 1.0, hdr: read("studio.hdr", encoding: none)),
)
```

IBL rotation

The `rotation` field spins the environment around the up axis (radians). Useful for aligning studio HDRs to your camera composition.

```
#render-gltf(helmet, read: R,
  camera: (2.5, 1.5, 2.5), up: (0, 1, 0), fov: 40,
  background: "#181820",
  ibl: (intensity: 1.2, rotation: 1.2),
)
```

Animations

Assets with animation channels get replayed at `time` seconds. `get-gltf-info` returns `max_animation_time` so you can build a scrub slider bounded to the actual clip length.

```
#render-gltf(fox,
  camera: (120, 90, 180), center: (0, 40, 0), up: (0, 1, 0),
fov: 40,
  background: "#1a1a22",
  ibl: (intensity: 1.3),
  time: 0.5,
  width: 45%,
)
```

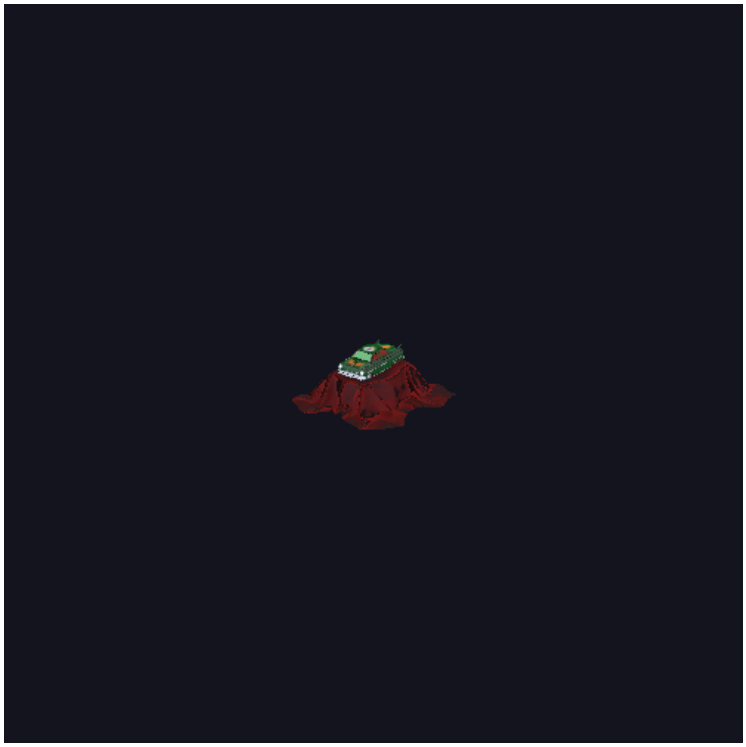
When the asset ships multiple clips (idle / walk / run / …), pick one with `animation_index`. `None` (the default) plays every clip stacked — last-write-wins per node channel — which is almost never what you want for multi-clip assets.

```
#render-gltf(cesiumman,
  camera: (2.5, 1, 2.5), center: (0, 0.9, 0), up: (0, 1, 0),
fov: 30,
  background: "#1a1a22",
  ibl: (intensity: 1.3),
  animation_index: 0,
  time: 0.3,
  width: 40%,
)
```

Material variants

`KHR_materials_variants` lets an asset ship multiple material sets. Pick one with `material_variant` (0-indexed).

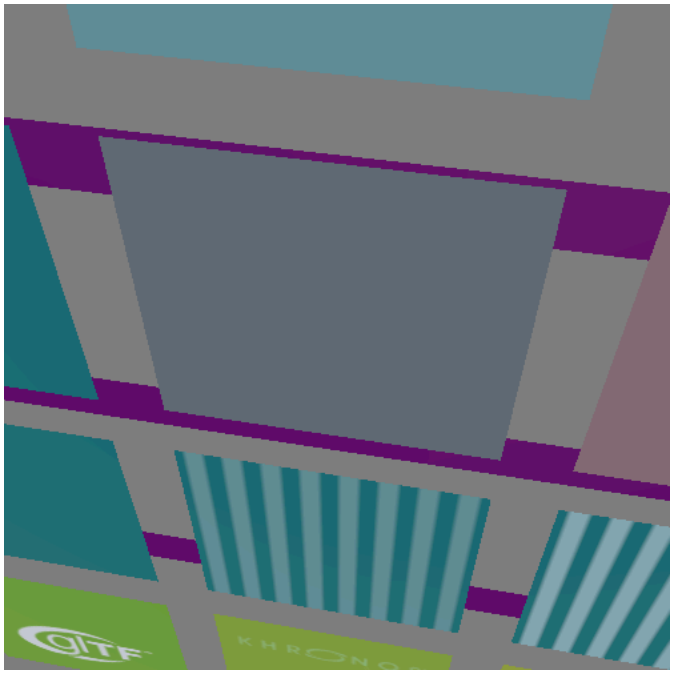
```
#render-gltf(toycar,
  camera: (0.4, 0.3, 0.55), center: (0, 0.05, 0), up: (0, 1, 0),
fov: 30,
  background: "#181820",
  ibl: (intensity: 1.4),
  material_variant: 0,
)
```



Diffuse transmission

`KHR_materials_diffuse_transmission` — matte back-lit lambertian. Common on cloth, leaves, paper. The example asset is a factor × color grid; the bottom rows show the green DT color activating as the factor increases.

```
#render-gltf(dt-test,
  camera: (1.5, 0.5, 1.5), center: (0, 0, 0), up: (0, 1, 0),
fov: 40,
  background: "#222",
  ibl: (intensity: 1.0),
  width: 90%,
)
```



Compression + quantization

The plugin decodes three common wire formats up-front, so downstream traversal never has to care.

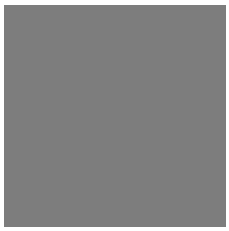
EXT_meshopt_compression

Emitted by `gltfpack` (Meshopt's own tool). Interleaved buffer decompression, transparent.

KHR_draco_mesh_compression

Google Draco. Decoded via `draco-oxide` (pure Rust). Up to 20× geometry compression on typical assets, no size penalty at render time.

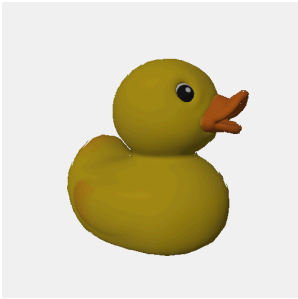
```
#render-gltf(box-draco, read: R,
  camera: (2, 1, 2), center: (0, 0, 0), up: (0, 1, 0), fov: 40,
  background: "#333",
  width: 30%,
)
```



KHR_mesh_quantization

`gltfpack` also emits quantized POSITION/NORMAL/TANGENT (i8/u8/i16/u16) paired with `KHR_texture_transform` for UV dequantization. Renders pixel-identically to the non-quantized asset — verified on the Duck below.

```
#render-gltf(duck-quant, read: R,
  camera: (2.5, 1.5, 2.5), center: (0, 0, 0), up: (0, 1, 0),
fov: 40,
  width: 40%,
)
```



Texture formats

Beyond the base PNG + JPEG:

EXT_texture_webp

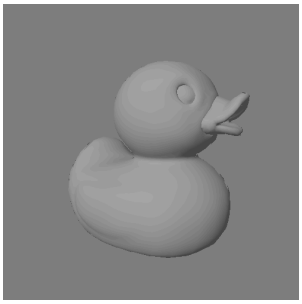
`image-webp` (pure Rust) handles both VP8 and VP8L variants. Second-most-common texture format in production glTF (Shopify AR, IKEA, Blender's default export in 3.4+).

```
#render-gltf(duck-webp, read: R,
  camera: (2.5, 1.5, 2.5), center: (0, 0, 0), up: (0, 1, 0),
fov: 40,
  width: 40%,
)
```

Unsupported formats

`KHR_texture_basisu` (KTX2 / Basis Universal) and `EXT_texture_avif` aren't decoded — they'd pull in 1 MB of decoder deps for formats we ship without today. Assets using them render with a **placeholder white texture** per material (geometry stays intact) rather than crashing. Below: the same Duck with KTX2 textures, silhouette-shaded because the base color reads as white:

```
#render-gltf(duck-ktx, read: R,
  camera: (2.5, 1.5, 2.5), center: (0, 0, 0), up: (0, 1, 0),
fov: 40,
  background: "#333",
  width: 40%,
)
```



Workaround: pre-transcode via [`gltf-transform`](https://gltf-transform.dev) to WebP or PNG.

Metadata

`get-gltf-info` returns a dict — bounding box, triangle count, animation length. Useful for driving auto-framing or a scrub slider without touching the render path.

```
#let info = get-gltf-info(helmet, read: R)
#{
  raw(block: true, lang: "yml",
    "triangles: " + str(info.triangles) + "\n" +
    "bbox_min:  [" + info.bbox_min.map(x => str(calc.round(x, digits: 2))).join(", ")
+ "]\n" +
    "bbox_max:  [" + info.bbox_max.map(x => str(calc.round(x, digits: 2))).join(", ")
+ "]\n" +
    "center:    [" + info.center.map(x => str(calc.round(x, digits: 2))).join(", ") +
"]\n" +
    "radius:    " + str(calc.round(info.radius, digits: 2)) + "\n" +
    "max_animation_time: " + str(info.max_animation_time)
  )
}
```

```
triangles: 15452
bbox_min:  [-0.95, -0.9, -1.19]
bbox_max:  [0.94, 0.9, 0.81]
center:    [0, 0, -0.19]
radius:    1.64
max_animation_time: 0
```

What's supported vs. not

Core spec: glTF 2.0 JSON + GLB (single-file and split with external `.bin`/textures), meshes (POINTS/LINES/TRIANGLES), skinning (JOINTS_0 + JOINTS_1, WEIGHTS_0 + WEIGHTS_1, up to 8 influences per vertex), morph targets, animations (TRS + morph weights, all three interpolation modes), multiple UV sets (TEXCOORD_0/1/2), vertex colors, cameras (perspective + orthographic), multiple scenes, multiple animations, textures with samplers, sparse accessors, KHR_mesh_quantization.

Rendering: Cook-Torrance GGX PBR, IBL from HDR (Radiance / RGBE), shadow maps (PCF + PCSS), alpha modes (OPAQUE / MASK / BLEND via WBOIT), double-sided normal flip, SSAO, FXAA, SSAA ×2/×4.

Extensions: `KHR_lights_punctual`, `KHR_materials_unlit`, `KHR_materials_transmission`, `KHR_materials_ior`, `KHR_materials_specular`, `KHR_materials_emissive_strength`, `KHR_materials_volume`, `KHR_materials_clearcoat`, `KHR_materials_sheen`, `KHR_materials_iridescence`, `KHR_materials_anisotropy`, `KHR_materials_dispersion`, `KHR_materials_diffuse_transmission`, `KHR_texture_transform`, `KHR_materials_pbrSpecularGlossiness`, `KHR_materials_variants`, `EXT_texture_webp`, `EXT_meshopt_compression`, `KHR_draco_mesh_compression`.

Not supported (assets render with graceful fallback — placeholder white texture or silent skip):

- `KHR_texture_basisu` (KTX2). Biggest gap — dominant production texture format.
- `EXT_texture_avif`. Rare in the wild.
- `KHR_animation_pointer`. Animations targeting material / light / camera properties via JSON pointer are ignored.
- TEXCOORD_N for N ≥ 3 (collapses to slot 2).
- Sparse accessors combined with quantization.

See `crates/maquette-gltf/README.md` for the full compliance matrix and per-extension notes.